GEOMETRY NODES PIPELINE

Meteor Facade v7 / B&W Image → Tiling Mesh

IMAGE ZONE SYSTEM

INPUTS
IMG_A + IMG_B
Two B&W images loaded in Shader Editor
Each defines one bit of a 2-bit zone index
IMG_A → bit 0 | **IMG_B** → bit 1

GEOMETRY NODES
UV Read
› InputNamedAttribute (FLOAT_VECTOR)
› Auto-detects active UV map name
› Outputs (U, V, 0) per mesh vertex

UV TRANSFORM
Scale + Offset
› VectorMath SCALE → UV Scale A / B
› VectorMath ADD → UV Offset A / B
› IMG_B gets +0.37, +0.61 decorrelation shift

GEOMETRYNODEIMAGETEXTURE × 2
Image Sampling
Sampled at transformed UV per vertex
Extension: **REPEAT** · Interpolation: **Linear**

```
bit0 = Color_A > Threshold
bit1 = Color_B > Threshold
```

BIT ENCODING
Zone Index 0 – 3
```
img_zone = bit0 + bit1 × 2
→ 0 / 1 / 2 / 3 (each ≈ 25 % surface)
```

HEIGHT GRADIENT
Gradient Mask
› World position projected along rotatable axis
› Normalised by Height Min / Max
› MapRange SMOOTHSTEP → gradient value
› TexNoise blur (Scale 12, Detail 2)
› use_img = gr_n > 0.5

ZONE MULTIPLIER
Effective Zone
```
eff_zone = img_zone × use_img
(zone 0 fills base when use_img = 0)
```
Per-face zone selection: |eff_zone – zi| < 0.5

INPUTS

ACTIVE OBJECT
Input Mesh
Must have UV unwrap (any UV map name)
Script auto-detects active UV layer
Any closed or open surface geometry

GEOMETRY NODES MODIFIER

CONTEXT

PREPROCESSING
Boundary SubdivideMesh
Level = **Boundary Detail** (0–5)
Increases vertex density at zone edges
Gives cleaner image-zone boundaries

PER-ZONE × 4

ZONE 0 | ZONE 1 | ZONE 2 | ZONE 3

Each zone runs the full pipeline
in parallel → 4 geometry streams

MERGE STREAMS
Join Geometry
All 4 zone meshes joined into one geometry

OUTPUT CONTROL

SWITCH NODE
Debug Maps Toggle
OFF → FULL PANELS | ON → ZONE PREVIEW

Debug: mesh colored R/G/B/Y per zone
using emission materials (no panels)

OUTPUT

MODIFIER OUTPUT
Tiling Facade Mesh
4 material zones, noise-extruded panels
Gap joints between each panel
Density controlled per zone via Grid Zi

Z0 SmoothStone | Z1 RoughStone
Z2 BlackGlass | Z3 MetalPanel

PER-ZONE PIPELINE ×4

MESH

ZONE SELECTION
Face Mask
Derived from **eff_zone** (left column)
Math: ABS(eff_zone – zi) < 0.5
Gives 0.0 / 1.0 per face

GEOMETRYNODESEPARATEGEOMETRY
Separate Zone Faces
→ Domain: FACE
→ Input: boundary-subdivided mesh
→ Output: only this zone's faces

GEOMETRYNODESUBDIVIDEMESH
Panel Grid Subdivision
→ Level = **Grid Zi** (0–5)
→ 4^N panels per original face
→ Independent per zone

Z0: 2 · Z1: 3 · Z2: 4 · Z3: 2 (defaults)

GEOMETRYNODESCALEELEMENTS
Gap Between Panels
→ Domain: FACE
→ Scale = 1 – Gap Size
→ Shrinks each face → visible joint

SHADERNODETEXNOISE + MATH
Per-Panel Extrude Depth
→ TexNoise 3D (Detail 4, Roughness 0.7)
→ Scale = **Height Noise**
→ Fac × Depth Zi → per-face offset

```
depth = noise.Fac × Depth Zi
```

GEOMETRYNODEEXTRUDEMESH
Extrude Panels
→ Mode: FACES
→ Individual: True
→ Offset Scale: depth (noise-driven)
→ Each face extruded independently

GEOMETRYNODESETMATERIAL
Assign Material
One Principled BSDF material per zone
Applied to entire zone geometry stream

Rough 0.95 | Rough 0.88 | Metal 0.04 | Metal 0.92

ZONE CONTROL
Image Parameters
UV Scale A / B — image tiling frequency
UV Offset A / B — shift image on surface
Threshold — zone balance (0.5 = equal)
Boundary Detail — zone edge smoothness (0–5)

GRADIENT CONTROL
Height Parameters
Height Min / Max — world Z range of building
Gradient Start / End — transition zone-0 to image
Gradient Rotation — tilt gradient direction

PANEL CONTROL
Tile Parameters
Grid Z0–Z3 — panel density per zone (4^N)
Gap Size — joint between panels (0–0.4)
Depth Z0–Z3 — max extrusion per zone
Height Noise — noise frequency for depth

● Mesh I/O ● Zone System ● Image Textures ● Gradient / Noise ● Per-Zone Pipeline ● Materials ● Output